Machine Learning | Assignment 1 | Aditya Mallakula

1. **Linear Regression with One Variables: (15 points):**
    a. **Implement linear regression to predict tumor size (Tumor Size) using the "Mean Texture" feature from the dataset. "Tumor Size" will be the target variable.**

I implemented linear regression using "Tumor Size" as the target variable and "Mean Texture" as the independent variable from the dataset. I split the data into training and test buckets with a 80:20 (Training : Test) ratio and calculated the theta values (Intercept and coefficient) using the Gradient Descent formula.

The theta1 & theta2 values are:
- Theta1 (m): 0.08750787925525076
- Theta2 (b): 0.9885482292593702

    b. **Evaluate performance using metrics (such as Mean Squared Error (MSE), R-squared (R^2), and Adjusted R-squared (Adjusted R^2). You may also use graphs for explaining your observations.**

With the theta values obtained, I calculated MSE & R-square for the model.
- MSE = 2.1229263984568165
- R-square = -0.12887518481467497

<u>My observation:</u> Though the cost of the model decreased over time and stabilised at 1.9613979284218892, the Mean Texture feature by itself is not enough to effectively predict the Tumor Size. The data is scattered unevenly and predictions have a lot of errors from the plot graph.
I was unable to compare the value of R-square as the value that showed up was negative. Due to this I can confirm that Mean Texture is a bad fit for predicting the Tumor Size.

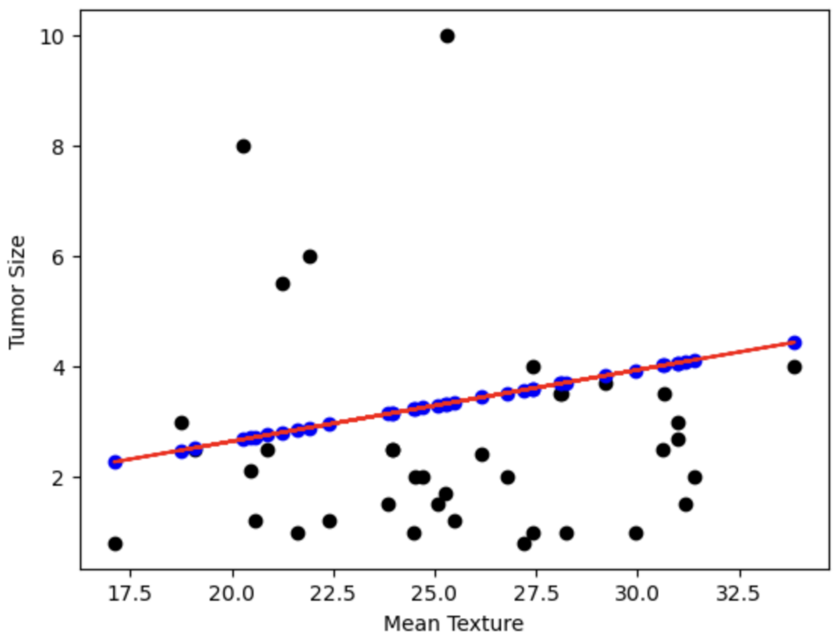

Fig1 - Graph shows the Predicted values among the scattered data values

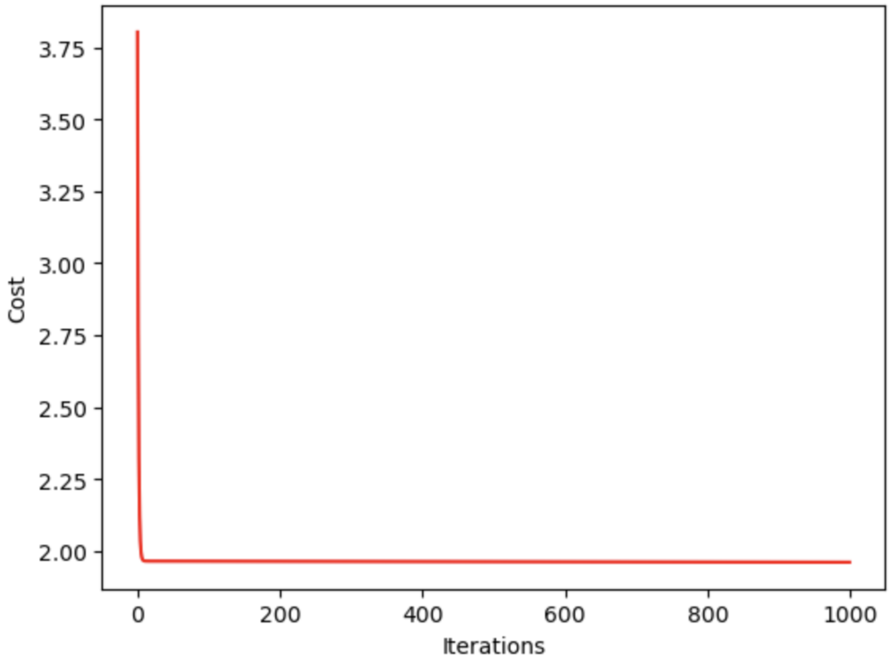

Fig2 - Graph shows the cost function being stabilised through multiple iterations

2. **Linear Regression with Two Variables: (15 points)**
    a. **Add 'Lymph Node Status' as an additional input feature to the previous linear regression model. Does adding this feature improve the performance of the model? Compare the performance of the models in Question 1 and Question 2 using evaluation metrics (such as Mean Squared Error (MSE), R-squared (R^2), and Adjusted R-squared (Adjusted R^2). You may also use graphs for explaining your observations.**

I have implemented linear regression on the dataset, split the data into training and test buckets and calculated the theta values (Intercepts and coefficient) using Gradient Descent formula.

The theta0, theta1 & theta2 values are 0.047683545150693486 0.2559879799888188 0.9606623863486354 respectively.

With the theta values obtained, I calculated MSE & R-square for the model.

- MSE = 1.840293116717243
- R-square = 0.021416270598224152
- BIC = 9.477555704583487

<u>My observation:</u> Compared to the previous model, the cost has decreased. But the change is not too drastic. However, when we compare the R-square values, there is a noticeable change. The current model has a better value in comparison. On comparing all provided data from both the models it is easy to say that this model performs better than the previous one. On the other hand, these are just 2 variables in the large data set and are still not enough to effectively predict the target value.

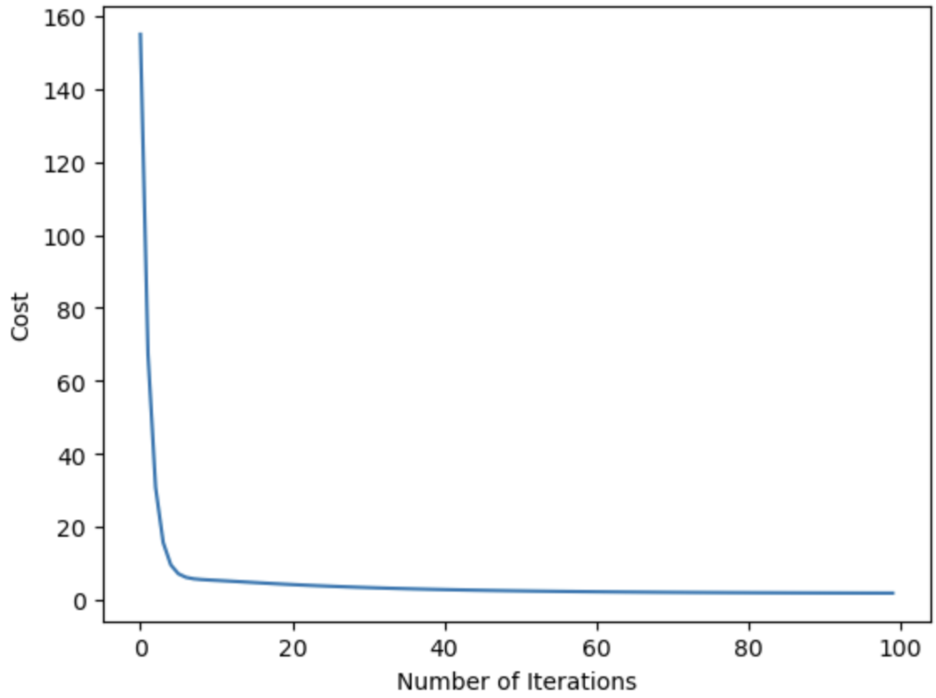

Fig3: The graph depicts the cost of the model over 100 iterations.

3. **Stepwise Linear Regression: (50 points)**
    a. **Implement forward stepwise linear regression with the chosen features. The process involves iteratively adding one feature at a time from your selection. After adding each feature, evaluate the model's performance using metrics such as Mean Squared Error, R-squared, Adjusted R-squared, or BIC-Bayesian Information Criterion (preferred). Choose the feature that contributes the most to improving the model's performance and add it to the model. Continue this iterative process for a total of 5 iterations. Explain your selection criteria for adding or removing features. Features: 'Mean Radius', 'Mean Perimeter', 'Mean Area', 'Mean Smoothness', 'Mean Symmetry', 'Mean Fractal Dimension', 'Worst Radius', 'Worst Area', 'Worst Symmetry', 'Lymph Node Status'**

I have implemented Forward stepwise linear regression using the features -'Mean Symmetry', 'Mean Fractal Dimension', 'Worst Radius', 'Worst Symmetry', 'Lymph Node Status'; from the dataset. I then split the data into training and test buckets in 80:20 (training : test) ratio and individually calculated the theta values (Intercepts and coefficient) for each feature using the Gradient Descent formula.
Once the theta values were obtained, I then used those values to calculate the MSE/Cost function and BIC of the model. The feature with the best BIC value was merged with the rest of the 4 features and another iteration of the above was performed on the model that was obtained. I continued this for a total of 5 iterations where I was able to get a final model with the best BIC value.

- The best model has the BIC value of 14.400277154338525 and contains Mean Fractal Dimension + Mean Symmetry + Worst Symmetry + Lymph Node Status.
- The minimum MSE for the model with the most effective combination of features is 0.04988670238029497

b. **For this task, you will be given a list of 10 features as below. Implement backward stepwise linear regression, beginning with a model that includes all 10 features. Remove one feature at a time from the model and evaluate the model's performance after each removal. Remove the feature that has the minimal adverse effect on the model's performance. Continue this iterative process for a total of 5 iterations. Please provide a detailed explanation of your criteria for selecting which features to remove or retain during this process. Features: 'Mean Radius', 'Mean Perimeter', 'Mean Area', 'Mean Smoothness', 'Mean Symmetry', 'Mean Fractal Dimension', 'Worst Radius', 'Worst Area', 'Worst Symmetry', 'Lymph Node Status'**

I have implemented backward stepwise linear regression using the features - 'Mean Radius', 'Mean Fractal Dimension', 'Worst Radius', 'Worst Symmetry', 'Lymph Node Status'; from the dataset. I then split the data into training and test buckets in 80:20 (training : test) ratio and individually calculated the theta values (Intercepts and coefficient) for all the features using the Gradient Descent formula.
Once the theta values were obtained, I then used those values to calculate the MSE/Cost function and BIC of the model. The feature with the least BIC value was removed from the bucket and another iteration of the above was performed on the model that was obtained. I continued this for a total of 5 iterations until I was able to get a final model with the best BIC value.

- The best model has the BIC value of 4.955648911495677 but contains just Mean Radius.
- Adding all the other features drastically decreased the BIC value and a negative value appeared for all the other models.
- The minimum MSE for the model with the most effective combination of features is 2.6389259095641378

c. **Compare the final model obtained from forward stepwise regression with the final model obtained from backward stepwise regression. Which one is better? Discuss the differences in terms of the selected features, model performance.**

On comparing the models using BIC & the MSEs, the forward stepwise regression (FSR) came up with the best model at the 4th Iteration with the best values respectively. Also, when compared to the forward stepwise model, the backward model functioned properly when there was only a single variable/feature. Whereas the forward model had 4 features. Hence I can confirm that the FSR approach performs better.

d. **Compare the performance of the model built using the features from Q.2 (a) with the resultant accuracies of the model built using the selected features Q.3(c). Which set of features performed better? Evaluate model performance using metrics like Mean Squared Error, R-squared, Adjusted R-squared or BIC-Bayesian Information Criterion (preferred). You may also use graphs for explaining your observation.**

In Q. 2(a) we used two features - Lymph Node Status & Mean Texture for evaluating the model. However, from the results of Q. 3(c), the resulting model had four features - Mean Fractal Dimension + Mean Symmetry + Worst Symmetry + Lymph Node Status.

Q. 2(a) BIC & MSE:

- MSE = 1.840293116717243
- BIC = 9.477555704583487

Q. 3(c) BIC & MSE:

- MSE = 0.04988670238029497
- BIC = 14.400277154338525

My Observation: On comparing the performance metrics of BIC & MSE, the model with two features does perform better. But, though the forward stepwise model has more features to evaluate, the numbers were really close to that of the model with two features. I can confirm that the Forward Stepwise regression model did perform better and is the most effective model of the two.

4. **Regularization and Feature Scaling: (20 points)**
    a. **For the best performing model in Q.3 (Model from Q.3(d)), does regularization improve the performance?**

On regularizing the best performing model from Q. 3, the cost of the new model is 0.03796059207071603. I have assigned the lambda value set as 1000. This is a slight decrease from the non regularized model, which was 0.04988670238029497.

The regularized coefficients are as follows:

- Before Regularization:
    - Coefficients = [1.0315048046338358, 1.0, 0.9986151734226889, 1.0000960757975619]
- After Regularization:
    - Coefficients = [1.4486733744179736, 1.448596013347552, 1.448550886611127, 1.4487179487179471, 1.0006224750099204]

The BIC values of the older model & newer model are as follows:

- Before Regularization:
    - BIC = 14.400277154338525
- After Regularization:
    - BIC = 15.30292580125751

Based on these parameters I feel that the performance of the model slightly increased since we added the lambda variable.

b. **Does Feature Scaling improve the performance for the model in Question 3(d)? Evaluate performance using metrics like Mean Squared Error, R-squared, Adjusted R-squared. You may also use graphs for explaining your observation.**

I scaled the feature using the formula <u>x - Mean(x) / Standard Deviation (x)</u>. After this I re-evaluated the Gradient Descent, Cost & BIC of the model.
Below is the comparison of both the models.

- Before Normalization:
  - MSE = 0.04988670238029497
  - BIC = 14.400277154338525

- After Normalization:
  - MSE = 0.04988239407491869
  - BIC = 14.396868848938109

<u>My Observation:</u> If we compare both the values, there is little to no change in the values of the normalized data. From this it is clear to state that there was no noticeable improvement to the model's overall performance.